

BioNode

The Omnivorous BioCNG Micro-Refinery

Bangladesh does not lack fuel. It lacks the wit to see fuel in what it throws away. The river chokes, the market rots, the field burns — **three sorrows, one answer.**

Every biogas plant before ours was built for one waste, and died when that waste stopped. Ours eats everything.

Six wastes. **One appetite.**

Three sources run all twelve months; the seasonal ones stack on top. The calendar never has a gap.

Kacha Bazar Waste

Urban wet markets, daily — we are paid to remove it

80–120 m³/t

Rice Straw

Boro & Aman, storable — otherwise burned in the field

200–300 m³/t

Water Hyacinth

Rivers & canals, free — clears the waterway

25–30 m³/t

Banana Pseudostem

Post-harvest farm residue, year-round

50–70 m³/t

Poultry & Dairy

Litter and dung, contracted — tipping fee

25–120 m³/t

Agro-Mill Waste

Press mud, fruit pulp, oil cake — mills pay us

60–150 m³/t

The gas is nearly free.

Tipping fees and biofertiliser cover roughly 90% of operating cost before a single cubic metre is sold. Our competitors buy their feedstock. We are paid to accept ours.

Blend Yard

C:N tuned to 25:1



Co-Digestion

AI-controlled



Clean-up

H₂S + CO₂



200-bar BioCNG

+ biofertiliser

BioNode

Built in Bangladesh. Fuel for the World.

Team BioNode · IIUC

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Why this works **where others failed.**

Mono-digestion is fragile: the wrong carbon-to-nitrogen ratio acidifies a reactor and it dies. Co-digestion is stable — but it needs a chemist to rebalance the recipe every time the input changes. At village scale, that chemist does not exist. **So we replaced the chemist with software.**

The ledger. **Three revenues, one machine.**

Modelled for Node-1 at 5 tonnes/day of blended feedstock. Design targets — not yet measured on a built unit.

FUEL

Tk 33 L

~200 m³/day of 200-bar BioCNG, per year

TIPPING FEES

Tk 9 L

Markets and mills pay us to take waste

BIOFERTILISER

Tk 15 L

Digestate dried and bagged for farms

PAYBACK

~2.7 yr

On ~Tk 80 L CAPEX · ~Tk 30 L EBITDA

The moat.

The chemist is the software

BioNode OS reads pH, VFA, temperature and gas quality, then re-tunes the blend in real time. Anyone can weld a tank; nobody else can run one on a diet that changes weekly.

The recipe library compounds

Every tonne digested teaches the model a better blend. Hardware is a week's work; ten thousand hours of blend telemetry is not.

Waste contracts are exclusive

A market signs one removal contract, not five. Each node locks its catchment — the supply moat closes behind us.

Honest about the machine. **Certain about the plan.**

DESIGNED — TODAY

Engineering & AI complete

Architecture, blend model, costed BOM, six supply routes mapped.

SUMMIT · JUL 26

Bench co-digestion

Multi-feedstock 50-L reactor with live control telemetry.

MVP · BY NOV 26

Full chain to a cylinder

Blend yard → digester → upgrade → 200 bar.

PILOT · 2027

Three Dhaka nodes

Gabtoli, Sayedabad, Mohakhali — audited field economics.

We are not selling gas.

We are building the operating system for Bangladesh's waste.